MODULE 37 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 2

Twenty fresh questions, same weighting, worked in full. Leans harder on recent changes.

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> An API's average response time sits comfortably at 180 milliseconds and the team's alarm never fires. Support tickets nevertheless report the application being unusably slow, and the affected users are consistently the same accounts.
>
> The alarm uses the Average statistic on the latency metric.

What should the team change?

A Reduce the alarm threshold below 180 milliseconds

B Alarm on a high percentile such as p99, which reveals the slow minority the mean conceals

C Switch the metric to high resolution

D Increase the evaluation periods so the alarm is less sensitive

◆ PAUSE AND DECIDE

What should the team change?

A Lower the threshold

Alarms on the same mean, which is healthy. It would fire constantly on normal traffic while still missing the slow tail.

B Alarm on p99

A mean hides a minority. If 2 percent of requests take 9 seconds, the average barely moves while those users have an unusable experience.

C High resolution

Changes how often the metric is published, not which statistic is evaluated. More frequent averages are still averages.

D More evaluation periods

Makes the alarm less sensitive, which is the opposite of the requirement. It already never fires.

THE SIGNAL

"Healthy average, unhappy minority" is the percentile signature. Resolution and statistic are different axes, and only one is the problem here.

> A team moves a high-volume log group to the Infrequent Access log class to reduce cost. Afterwards, an existing metric filter on that log group stops producing data and its alarm goes to INSUFFICIENT_DATA.
>
> Logs Insights queries against the log group still return results.

What is the cause?

A The metric filter must be recreated after a log class change

B Infrequent Access retains logs for a shorter period

C The Infrequent Access log class does not support metric filters

D The alarm needs its missing data treatment changed

◆ PAUSE AND DECIDE

What is the cause?

A **Recreate the filter**

Would imply the capability exists and merely needs re-establishing. The capability is absent in this log class.

B **Shorter retention**

Retention is configured independently of log class, and the scenario confirms queries still return data.

C **IA does not support metric filters**

The Infrequent Access class trades features for cost: no metric filters, no subscription filters, no Live Tail, no anomaly detection.

D **Missing data treatment**

Changes how the alarm interprets gaps. It cannot make a metric exist that is no longer produced.

THE SIGNAL

The log class is chosen at creation and trades capability for cost. Logs Insights still working is the clue that the data is fine and the feature is not.

> An EventBridge rule invokes a Lambda function. During a two-hour outage of a downstream API the function failed every invocation, and those events are now lost. The team wants both to capture failures for reprocessing and to replay a range of past events after a code fix.
>
> No dead-letter queue is configured on the rule's target.

What should the team configure?

A. A dead-letter queue only, which also allows replay

B. An EventBridge archive only, which also captures failures

C. Increase the retry attempts on the rule

D. A dead-letter queue for failed deliveries, and an event archive for replay

◆ PAUSE AND DECIDE

What should the team configure?

A DLQ only

Captures events EventBridge could not deliver. It holds nothing about events delivered successfully, so there is nothing to replay.

B Archive only

Stores matched events for replay. It does not capture delivery failures, which is the other half of the requirement.

C More retries

EventBridge already retries for up to 24 hours. It captures nothing and replays nothing.

D Both, for different jobs

A DLQ captures what could not be delivered; an archive stores what matched, so a range can be replayed later. Two requirements, two mechanisms.

THE SIGNAL

Two stated requirements, and each of A and B answers exactly one. Counting the requirements before reading the options settles this immediately.

> A database password must change every 30 days, in the database and in the configuration the application reads, without downtime and without human involvement. The value is currently a SecureString in Parameter Store with a policy that expires it after 30 days.
>
> Applications fail when the parameter expires.

What should the team do?

A. Move the credential to Secrets Manager and enable rotation, which changes it in the database and the secret together

B. Change the Parameter Store policy from expire to notify

C. Use an Advanced tier parameter, which supports rotation

D. Increase the expiry period to 90 days

◆ PAUSE AND DECIDE

What should the team do?

A Secrets Manager with rotation

Rotation changes the credential in the database and in the secret atomically. The alternating-user strategy gives rotation with no downtime.

B Expire to notify

Stops the outage by no longer deleting the value, and nothing then changes the password. The requirement is rotation.

C Advanced tier parameter

Advanced tier adds size, count and parameter policies. It does not add rotation; that is the distinction being tested.

D Longer expiry

Delays the same failure by 60 days, and still deletes the value rather than replacing it.

THE SIGNAL

A parameter policy expires or notifies. Only Secrets Manager rotates, meaning it changes the credential in two places at once.

A security group rule permitting inbound 0.0.0.0/0 on port 22 is discovered on a production instance. The team needs to know who made the change and when, and separately needs to see what the security group's configuration looked like before and after.

Both CloudTrail and AWS Config are enabled.

Which service answers which question?

A CloudTrail answers both, since it records the API call and its parameters

B Config answers both, since it records configuration over time

C CloudTrail identifies who and when; Config shows the before and after configuration

D Neither; VPC Flow Logs are required

◆ PAUSE AND DECIDE

Which service answers which question?

A CloudTrail answers both

It records the call and its parameters, so it shows the change requested. It does not maintain the resource's state over time.

B Config answers both

Config records configuration items and relationships. It does not record the identity that made a call.

C CloudTrail for who; Config for what

Complementary by design: CloudTrail records the API call and the caller, Config records the resulting resource state and its history.

D Flow logs needed

Flow logs record traffic that flowed, not configuration changes. The wrong category of evidence entirely.

THE SIGNAL

Who did it, what it looks like now, how it is performing: CloudTrail, Config, CloudWatch. The trio that appears in some form on almost every attempt.

A retail site sees traffic rise from a quiet overnight baseline to five times that level, between 08:00 and 08:20 every weekday. The pattern is unchanged for eighteen months. Instances take about seven minutes to become useful, and target tracking on CPU is configured.

The Auto Scaling group is nowhere near its maximum capacity.

What should be added?

A Predictive scaling in ForecastAndScale mode, retaining target tracking

B A lower target tracking threshold

C A step scaling policy with larger steps

D A higher maximum capacity

◆ PAUSE AND DECIDE

What should be added?

A Predictive plus target tracking

Eighteen months of consistent history is exactly what a forecast needs, and capacity arrives before 08:00 rather than seven minutes after demand.

B Lower threshold

Reacts sooner and still only after load arrives, then runs excess capacity for the other twenty-two hours.

C Larger steps

Still reactive. However aggressive the steps, instances take seven minutes against a twenty-minute ramp.

D Higher maximum

The scenario states the group is nowhere near its maximum, so the ceiling is explicitly not the constraint.

THE SIGNAL

Every reactive answer loses the same race: seven minutes to boot against a twenty-minute ramp. Only scaling before 08:00 wins it.

> An RDS for MySQL database is deployed as Multi-AZ. Reporting queries are slowing the application, and a developer proposes directing the reporting tool at the standby instance to spread the load.
>
> Availability requirements are already met.

What is wrong with the proposal, and what should be done?

A. Nothing is wrong; the standby can serve read traffic

B. The standby is in another Region, so latency would be too high

C. Multi-AZ must be disabled before a read replica can be added

D. The Multi-AZ standby serves no traffic and has no endpoint. Add a read replica for the reporting workload

◆ PAUSE AND DECIDE

What is wrong with the proposal, and what should be done?

A The standby can serve reads

It cannot. A classic Multi-AZ standby is invisible: no endpoint, no reads, and no backups taken from it.

B Another Region

Multi-AZ is within one Region by definition. Cross-Region means a read replica or a global database.

C Disable Multi-AZ first

Invented requirement. Read replicas and Multi-AZ coexist, and most production designs use both.

D Standby serves nothing; add a replica

Multi-AZ is availability, replicas are scale. Different mechanisms answering different requirements.

THE SIGNAL

Availability versus scale, synchronous versus asynchronous, automatic versus manual. The most confused pair in the RDS syllabus.

> An engineer deletes a production RDS instance in error. Automated backups were enabled with 14-day retention. The 'retain automated backups' option was not selected during deletion, and no manual snapshots exist.
>
> The team asks how to restore to a point five minutes before the deletion.

What is the situation?

A Restore to the required point in time from the retained automated backups

B The automated backups were deleted with the instance, so no restore is possible

C Restore from the final snapshot that RDS always takes automatically

D Recover the instance from the RDS recycle bin within 14 days

◆ PAUSE AND DECIDE

What is the situation?

A Restore from automated backups

There are none. Automated backups are a property of the instance and are removed with it unless retention was explicitly requested.

B Nothing to restore from

Uncomfortable and correct. This is precisely why manual snapshots or AWS Backup vault copies exist as a separate control.

C The automatic final snapshot

The most tempting option. A final snapshot is offered at deletion, not taken automatically, and the scenario says it was not selected.

D RDS recycle bin

Recycle Bin covers EBS snapshots and AMIs, not RDS instances. Invented behaviour for this service.

THE SIGNAL

Some questions have an unpleasant correct answer. Recognising that a recovery is impossible is a real skill, and the exam does test it.

A team configures Route 53 failover routing between two Regions. The primary record's health check targets an internal Application Load Balancer in a private subnet. The health check has never once reported healthy, so all traffic goes to the secondary Region.

The load balancer and its targets are confirmed healthy.

What is the cause and the fix?

A The failure threshold is set too low; raise it

B Failover routing requires a calculated health check

C Endpoint health checkers run on the public internet and cannot reach a private endpoint. Use a CloudWatch alarm health check instead

D The record's TTL is too high

◆ PAUSE AND DECIDE

What is the cause and the fix?

A Threshold too low

Would cause flapping on intermittent problems, not a check that has never once succeeded since creation.

B Calculated check required

Invented requirement. Failover routing works with any health check type, including none at all.

C Public checkers cannot reach a private endpoint

Endpoint health checks originate on the public internet. An internal load balancer has no public address, so every probe fails.

D TTL too high

TTL governs how long clients cache an answer. It has no bearing on whether a health check succeeds.

THE SIGNAL

"Never once healthy" is the signal. Intermittent problems flap; a structural impossibility fails from the very first probe.

A CloudFormation stack creates an Auto Scaling group of application servers. The stack reports CREATE_COMPLETE about three minutes after launch, but the application takes roughly nine minutes to install and start on each instance. A pipeline runs smoke tests as soon as the stack completes, and they fail unpredictably.

The team wants CREATE_COMPLETE to mean the application is genuinely serving.

What should be added?

A. A DependsOn attribute between the Auto Scaling group and the load balancer

B. A CreationPolicy with a ResourceSignal, and cfn-signal at the end of user data

C. An UpdatePolicy with AutoScalingRollingUpdate

D. A fixed 10-minute wait step in the pipeline

◆ PAUSE AND DECIDE

What should be added?

A DependsOn

Changes ordering between resources. The problem is not order; it is that an instance counts as created before it is ready.

B CreationPolicy plus cfn-signal

The stack waits for each instance to signal success from its own user data, so CREATE_COMPLETE means the application is actually running.

C UpdatePolicy

Governs how updates replace instances. The scenario describes the initial creation reporting success too early.

D A fixed wait

The tempting shortcut. It waits whether or not anything worked, so a failed installation still reports success ten minutes later.

THE SIGNAL

Launched is not ready. CreationPolicy is the only mechanism that makes CloudFormation's definition of complete match the application's.

A team is designing blue/green deployments for an ECS service in 2026. They need a bake period, automated validation against the new version before customers reach it, and automatic rollback on a CloudWatch alarm. They have no existing CodeDeploy configuration.

The service runs on Fargate behind an Application Load Balancer.

What should they implement?

A ECS native blue/green, setting the deployment controller to ECS, with a test listener, bake time and alarm rollback

B ECS blue/green through CodeDeploy, with an appspec file and a deployment group

C An ECS rolling update with minimum healthy percent set to 100

D Two ECS clusters with Route 53 weighted records between them

◆ PAUSE AND DECIDE

What should they implement?

A ECS native blue/green

Built into ECS since July 2025, with bake time, a test listener for pre-customer validation, and alarm rollback. AWS names it the default choice for new deployments.

B CodeDeploy

Works, and since July 2025 it is extra machinery: an application, deployment group, appspec and role. The scenario removes the one good reason to choose it.

C Rolling at 100 percent

Avoids the capacity dip and discards old tasks as it goes, so rollback means another full deployment. It offers no test listener.

D Two clusters plus DNS

Achieves blue/green at the DNS layer, where TTLs and resolver caching make rollback slow and imprecise.

THE SIGNAL

Every course written before mid-2025 says ECS blue/green requires CodeDeploy. It does not, and AWS no longer recommends that path for new work.

A team is configuring Terraform remote state for a new AWS project. A colleague provides a runbook from an earlier project. It specifies an S3 bucket for state, plus a DynamoDB table with a LockID partition key for locking.

The team wants the configuration currently recommended.

What should the team do?

A Follow the runbook exactly, since it is a proven pattern

B Store state locally and commit it to the repository

C Use the DynamoDB table but drop the S3 bucket

D Use an S3 bucket with use_lockfile enabled, and no DynamoDB table

◆ PAUSE AND DECIDE

What should the team do?

A. Follow the runbook

Works, and DynamoDB locking is deprecated in HashiCorp's own documentation and slated for removal. A runbook that has aged.

B. Local state in Git

State contains resource identifiers and often secrets, and committing it provides no locking. Two people applying at once corrupt it.

C. DynamoDB without S3

DynamoDB was only ever the lock, never the state store. Removing the bucket removes the state itself.

D. S3 with use_lockfile

S3 native locking is the current mechanism: a .tflock object beside the state file, one fewer resource, and no deprecated arguments.

THE SIGNAL

The same shape as Incident Manager and ARC readiness check: an experienced colleague's advice that has quietly gone out of date.

A Lambda function is triggered asynchronously by S3 events. About 3 percent of objects cause the function to fail permanently because of a malformed file. The team discovers this weeks later and has no record of which objects failed.

The function has no destination and no dead-letter queue configured.

What explains the missing record, and what should be configured?

A S3 event notifications do not retry, so the events were never delivered

B The function timeout is too short; increase it

C Asynchronous invocation retries twice and then discards the event silently. Configure an on-failure destination

D Enable S3 versioning to preserve the failed objects

◆ PAUSE AND DECIDE

What explains the missing record, and what should be configured?

A S3 does not retry

Asynchronous Lambda invocation does retry, twice. The problem is what happens after the retries are exhausted.

B Timeout too short

The failures are permanent and caused by malformed input. A longer timeout produces the same failure more slowly.

C Silent discard; add a destination

With no destination and no DLQ, a permanently failing async event is dropped and the only trace is a metric. A destination captures request and response context.

D Versioning

Protects objects that were never at risk. It says nothing about which invocations failed.

THE SIGNAL

Destinations superseded the DLQ for asynchronous invocation: they capture request and response context and support on-success as well.

A platform team must let application teams create IAM roles for their own workloads. Those roles must never modify IAM, delete CloudTrail trails, or act outside approved Regions. The platform team will not review each role by hand.

Application teams must retain the ability to create roles.

What should the platform team implement?

A. A permissions boundary attached to the application teams' own roles

B. An SCP denying iam:CreateRole in the application accounts

C. An AWS Config rule detecting over-permissive roles after creation

D. A permissions boundary, plus a condition on the teams' policies requiring that boundary on any role they create

◆ PAUSE AND DECIDE

What should the platform team implement?

A Boundary on their own roles

The near miss: right feature, wrong attachment point. It caps what the teams themselves can do, not the roles they create.

B Deny CreateRole

Removes the delegation entirely. The requirement is that teams should be able to create roles, safely.

C Config rule

Detective, so an over-permissive role exists until someone acts. The requirement is that it never can.

D Boundary plus the required-boundary condition

The condition forces every role they create to carry the boundary, and the boundary caps what those roles can ever do. Delegation with a guarantee.

THE SIGNAL

The boundary is not the control. The condition requiring the boundary is. Without it, teams simply create a role with no boundary at all.

A regulator requires that seven years of transaction records held in S3 cannot be deleted or altered by anyone in the organisation for the full retention period. The requirement explicitly names administrators and the account root user.

The bucket already has versioning enabled.

What should be implemented?

A S3 Object Lock in compliance mode with a seven-year retention period

B S3 Object Lock in governance mode with a seven-year retention period

C MFA delete on the bucket

D A bucket policy denying s3:DeleteObject to all principals

◆ PAUSE AND DECIDE

What should be implemented?

A Compliance mode

The only control here that the root user cannot override. Retention cannot be shortened and objects cannot be deleted until it expires.

B Governance mode

Designed to be overridable by a user holding s3:BypassGovernanceRetention. The requirement says nobody, which rules it out.

C MFA delete

Raises the bar without removing the capability. An administrator holding the MFA device can still delete.

D Bucket policy

Whoever can write bucket policies can rewrite this one. A policy cannot bind the people who control policies.

THE SIGNAL

"Including the root user" eliminates every control a privileged person can reverse. The same discriminator applies to Vault Lock and CloudTrail.

An audit finds several S3 buckets across member accounts whose bucket policies grant read access to named external AWS accounts. Application teams own their own buckets and the security team cannot review every policy. They need a central control preventing any principal outside the organisation from accessing any S3 bucket.

S3 Block Public Access is already enabled account-wide everywhere.

What should the security team implement?

A An SCP denying s3:PutBucketPolicy in all member accounts

B A resource control policy denying S3 access to principals outside the organisation

C Rely on Block Public Access, which already covers this

D An IAM Access Analyzer external access analyzer at organisation level

◆ PAUSE AND DECIDE

What should the security team implement?

A Deny PutBucketPolicy

Stops new bucket policies being written and leaves every existing grant intact. It also blocks legitimate use entirely.

B RCP denying external principals

An RCP caps what may be done to your resources, overriding any bucket policy, for every bucket at once. Exactly the data perimeter use case.

C Block Public Access

The sharp one. Access granted to a named external AWS account is not public access, so BPA permits it, and it is already enabled here.

D External access analyzer

Detective. It reports the sharing the audit has already found, and prevents nothing.

THE SIGNAL

Cross-account is not the same as public. Block Public Access is already enabled in this scenario and has done nothing about these grants.

An architecture spans three Availability Zones with EC2 instances in private subnets in each, an Auto Scaling group and an ALB across all three, and an RDS Multi-AZ database. A single NAT gateway sits in the first Availability Zone and all three private subnets route to it.

During a simulated failure of the first AZ, instances in the other two lose outbound internet access.

What should be changed?

A Add a second ALB in another Availability Zone

B Deploy a NAT gateway in each Availability Zone, with each private subnet routing to its local one

C Convert the database to a Multi-AZ DB cluster

D Enable cross-zone load balancing on the ALB

◆ PAUSE AND DECIDE

What should be changed?

A — A second ALB

An ALB already spans the zones you enable. The load balancer was never the failing component.

B — A NAT gateway per AZ

A NAT gateway is a zonal resource. Routing every private subnet through one makes that zone a single point of failure for all outbound traffic.

C — Multi-AZ DB cluster

Improves database failover time. The database was already Multi-AZ and is not what broke.

D — Cross-zone load balancing

Always on for an ALB, and it concerns inbound distribution rather than outbound internet access.

THE SIGNAL

The scenario lists four components correctly spread across zones, then names one that is not. That asymmetry is the whole question.

A CloudFront distribution in front of a dynamic web application reports a cache hit ratio of about 3 percent, and the origin is heavily loaded. The cache behaviour forwards all cookies and all query strings and includes them in the cache key. Most responses are identical for all users.

The application requires the CloudFront-Viewer-Country header for analytics.

What should the team do?

A Increase the maximum TTL on the behaviour

B Enable Origin Shield

C Add more origins to an origin group

D Reduce the cache key to what genuinely varies the response, and use an origin request policy to forward the country header

◆ PAUSE AND DECIDE

What should the team do?

A. Raise the maximum TTL

Objects are not expiring; they are never reused, because each request produces a distinct cache key. A longer TTL on a unique object achieves nothing.

B. Origin Shield

Collapses duplicate origin fetches for the same object. With a near-unique key per request, there are no duplicates to collapse.

C. More origins

Spreads the same load across more servers while the cache continues to do nothing.

D. Shrink the key; forward via origin request policy

Cookies in the cache key give every user their own copy. An origin request policy forwards the header without fragmenting the cache.

THE SIGNAL

A poor cache hit ratio is a cache key problem, not a TTL problem. Option A is the reflex answer and addresses expiry rather than uniqueness.

> A team must point example.com, the bare domain with no subdomain, at an Application Load Balancer. Route 53 rejects a CNAME record created for example.com. The identical CNAME created for www.example.com works correctly.
>
> The load balancer's IP addresses are managed by AWS and change over time.

What should the team create?

A. An alias record for example.com targeting the load balancer

B. An A record containing the load balancer's current IP addresses

C. A CNAME, after deleting the zone's SOA record

D. An NS record delegating example.com to the load balancer

◆ PAUSE AND DECIDE

What should the team create?

A An alias record

Alias is Route 53's extension that works at the zone apex, follows the load balancer's changing addresses, and is free for AWS targets.

B A record with IPs

The scenario states the addresses change. This works until AWS replaces a node, then fails silently for everyone.

C Delete the SOA record

Would break the zone entirely. SOA is mandatory, and the DNS standard forbids a CNAME coexisting with other records at the apex.

D NS delegation

Delegates authority for the zone to other name servers. A load balancer is not a name server.

THE SIGNAL

The comparison is supplied: www works, the apex does not. That is the CNAME-at-apex rule, and only an alias record solves it.

> An application in VPC A cannot reach a database in VPC B across a Transit Gateway. The database's security group permits the database port from the application's security group. That identical rule works correctly for application instances inside VPC B.
>
> Route tables and network ACLs have been verified as correct.

What is the most likely cause?

A Security group references never work across VPCs

B The Transit Gateway needs a NAT gateway on the path

C The Transit Gateway discards security group information unless referencing support is enabled on the attachment

D The database security group must use the application's CIDR range instead

◆ PAUSE AND DECIDE

What is the most likely cause?

A References never work across VPCs

Too absolute. They work across peered VPCs in the same Region, and across a Transit Gateway once referencing support is enabled.

B NAT gateway on the path

NAT is for internet access. Both VPCs are private and routing has already been verified.

C Referencing support not enabled

By default a Transit Gateway discards security group information about forwarded traffic, so a group-referencing rule has nothing to match against.

D Use a CIDR range

A workaround that would work, at the cost of the maintainable rule. The question asks for the cause, not a way around it.

THE SIGNAL

The controlled comparison is the clue: the identical rule works inside VPC B and fails across the Transit Gateway. Only the path differs.

How this set was weighted

- **Five Domain 1, four Domain 2, four Domain 3, three Domain 4, four Domain 5**, matching the real exam's proportions.
- **Four questions turned on recent changes**: the IA log class, ECS native blue/green, Terraform state locking, and resource control policies.
- **Three questions contained a controlled comparison**: something that works in one place and fails in another. That is always the clue.
- **One question had an unpleasant correct answer**: no restore was possible. The exam does ask you to recognise that.

> ON THE EXAM
>
> Compare your Domain 1 to 5 breakdown with practice exam 1. A domain weak in both is where your remaining revision belongs.